SCIENCE EXPLORER / GRADE 8

Force and Energy

Small pushes. Big discoveries.

Five visual missions, an original revision guide and 100 practice questions.

Original teaching material informed by KICD's revised 2024 Grade 8 curriculum. Independent prototype, not KICD-endorsed; teacher review required. Five introductions to strand 3.0, whose two sub-strands allocate 20 and 16 lessons. The original curriculum is linked, never bundled.

KICD Integrated Science Grade 8, revised 2024, strand 3.0: 3.1 Transformation of Energy and 3.2 Pressure. Printed curriculum pages 16-20; source viewer pages 28-32 of 37, with assessment rubrics on pages 33-34. Original teaching text, questions and animations, not an official assessment or a replacement for the full 36 lessons.

This guide is original teaching material, not a reproduction or redistribution of the KICD curriculum. The original source is available through the official listing and viewer links below.

Official listing: https://kicd.ac.ke/cbc-materials/curriculum-designs/grade-eight-designs/#category10

Original curriculum viewer: https://drive.google.com/file/d/1RpqZRmeMywbc1Tya0_x-tNJBs75Gy0-v/view

How to use: try the HTML animations, predict the answer, then open the optional explanation. Use this guide for deeper reading and practice. Check answers only after attempting questions.

Safety: use the virtual models only. No mains electricity, flames, heights, sharp tools, pressurised containers, mouth siphoning or vehicle experiments. A teacher must approve and supervise real practical work.

Your learning route

1. Energy forms & sources: Recognise energy forms and distinguish renewable from non-renewable sources.

2. Follow the energy: Explain simple energy transformations and account for less useful transfers.

3. Pressure on solids: Explore how force and contact area affect pressure.

4. Pressure in liquids: Describe qualitatively how pressure in a still liquid changes with depth.

5. Useful science, safer choices: Connect energy and pressure to devices, everyday life and safety.

Coverage: the first two missions introduce forms, sources and transformations of energy. The next two explore solid and liquid pressure. The final mission links applications and safety. This compact introduction does not replace the curriculum's practical investigations or full teaching time.

1. Energy forms & sources

Recognise energy forms and distinguish renewable from non-renewable sources.

Energy helps us explain changes and transfers. Moving objects have kinetic energy. A raised object has gravitational potential energy. Energy can also be stored chemically, elastically or in nuclear systems, and transferred electrically, by heating, by light and by sound.

An energy source is not the same as a form of energy. Sunlight, wind, falling water, geothermal resources and fuels can supply energy. Electricity is an energy-transfer pathway, not automatically a renewable primary source.

Renewable resources are replenished naturally on a human timescale. Examples include sunlight, wind and suitably managed flowing-water resources. Biomass can be renewable when regrowth keeps pace with use; cutting forests faster than they regrow is not sustainable.

Coal, petroleum, natural gas and uranium are non-renewable resources: available supplies are finite or form far more slowly than we use them. Nuclear energy does not require burning a fossil fuel, but mined uranium is still non-renewable.

Renewable does not mean impact-free or always available. Compare reliability, location, land use, pollution, costs and community needs. Our animations classify examples; they do not model an entire power station.

Visual mission

Where does the energy come from?

A source supplies energy. Renewable resources replenish on a human timescale.

Three simplified examples, not full power-station models. Renewable does not mean impact-free or always available.

Think it through

Choose two local energy sources. Explain how each is replenished, one use and one limitation.

2. Follow the energy

Explain simple energy transformations and account for less useful transfers.

Energy is conserved: it is not created or destroyed by an ordinary transformation. Track where it comes from and where it goes. Energy described as 'wasted' is usually dispersed into less useful stores, often warming the surroundings.

In a battery-powered torch, chemical energy supplies an electrical transfer to the lamp. The lamp emits light and also transfers energy by heating. A lamp does not convert every input into useful light.

For a ball falling from rest, gravitational potential energy decreases and kinetic energy increases. In the ideal animation, air resistance is ignored, so total modelled mechanical energy stays constant until just before impact. A real fall also transfers energy to the air.

A motor uses an electrical transfer to produce movement; a generator uses movement to supply an electrical transfer. A hydroelectric scheme uses falling water to drive a turbine and generator. A solar photovoltaic panel converts light into an electrical supply.

Rubbing hands transfers energy through friction and warms them. A microphone converts sound-related vibrations into an electrical signal. Real devices commonly have several outputs; a simple arrow chain is a model, not the complete engineering.

Visual mission

Energy does not disappear

Follow the transfers: account for the useful output and the surroundings.

The falling-ball bars show proportions of modelled mechanical energy before impact, ignoring air resistance. The torch is a separate energy-chain model.

Think it through

Draw an energy chain for a device you use. Include one useful output and one less useful transfer.

3. Pressure on solids

Explore how force and contact area affect pressure.

Pressure describes normal force per unit contact area. Normal means perpendicular to the surface. Increasing downward force over the same contact area raises pressure. Spreading the same force over a larger contact area lowers pressure.

Weight is a gravitational force, measured in newtons, while mass is measured in kilograms. For an object resting on a horizontal support in a simple balanced situation, its weight helps determine the contact force.

The virtual model uses pressure = force divided by area. With force in newtons and area in square centimetres, multiply the numerical ratio by ten to express pressure in kilopascals. For example, 40 N on 40 square cm gives 10 kPa.

Wide straps distribute a load over more area. Broad tyres and foundations can reduce pressure on soft ground. A thin cutting edge concentrates a force over a small area; sharp tools are examples to discuss, not unsupervised activities.

The model assumes uniform contact and a perpendicular force. Real materials deform, so actual pressure may not be even across a surface. Change one slider at a time to see a fair comparison.

Visual mission

Same push. Different footprint.

More force raises pressure; more area lowers it, with the other variable fixed.

Uniform average pressure model with a perpendicular force. Square cm are converted to square metres for the kPa result. The moving dot shows force direction, not a changing numerical force.

Think it through

Keep force fixed, then double the contact area. Predict and explain what happens to pressure.

4. Pressure in liquids

Describe qualitatively how pressure in a still liquid changes with depth.

In the same still liquid, pressure due to the liquid increases with depth below the surface. Deeper points have more liquid above them. This lesson treats the relationship qualitatively; no liquid-pressure calculation is required.

At equal depth in the same connected liquid at rest, pressure is equal, even if the container has an unusual shape. Depth means vertical distance below the liquid surface, not distance along the wall.

Liquid pressure acts in all directions at a point. It is not simply a downward push. The virtual probes are at the same depth and show the same pressure due to the water.

The model keeps liquid type and gravity fixed and shows liquid pressure relative to the surface; the surrounding air pressure is not added to the display. Different liquid densities can also affect pressure at a given depth.

Tanks and dams must withstand greater liquid pressure at greater depths, among other engineering demands. A simulation avoids holes, spills, diving or pressurised containers. Do not copy a pressure demonstration without a teacher.

Visual mission

Go deeper. Feel more pressure.

Greater depth means greater liquid pressure. Equal depth means equal pressure under the same conditions.

Qualitative model of one still liquid. Both probes stay at equal depth. The bar shows pressure due to the water relative to the surface, excluding air pressure; it is not calibrated in pascals.

Think it through

Explain why two probes at the same depth read equally, and predict what changes when both move deeper.

5. Useful science, safer choices

Connect energy and pressure to devices, everyday life and safety.

In a simple hydraulic system, a change of pressure is transmitted through a confined liquid. Brake fluid can transfer a pedal input to a vehicle's brake mechanism. Friction at the brakes transfers some motion-related energy into thermal stores as the vehicle slows.

A syringe can move liquid when the plunger changes pressure. Hydraulic brakes use liquids, whereas bicycle pumps compress gas. Drinking through a straw involves a pressure difference: lower pressure in the mouth and straw allows atmospheric pressure on the drink to push liquid up.

A siphon can move liquid from a higher reservoir to a lower outlet when the system is suitably filled and arranged. Gravity and pressure differences matter; it is not a source of endless energy. Never start a siphon by mouth, especially with fuel or unknown liquids.

Stay away from mains wiring, hot devices, exposed moving machinery and pressurised containers. Do not look at the Sun or bright lasers, and protect hearing from excessive sound. Never burn fuels or run engines indoors because harmful gases can accumulate.

Seat belts and helmets help reduce injury risk; a learning animation does not replace correct use or road-safety rules. Real hydraulic systems, electrical devices, sharp tools and vehicles need trained supervision. Share explanations and source links, not unsafe experiments or personal student data.

Visual mission

A push becomes a safer stop

Brake fluid transmits pressure; friction transfers motion-related energy into thermal stores.

A conceptual chain, not a diagram for building or repairing brakes. Motion is illustrative, not a computed vehicle stopping distance.

Think it through

Choose one device: a brake, syringe, pump or straw. Explain the pressure idea, any energy transfers, and a safety precaution.

Revision cards

Kinetic energy

Energy associated with motion.

Gravitational potential energy

Energy stored because of position in a gravitational field, relative to a reference level.

Renewable source

A resource replenished on a human timescale; sustainable management can matter.

Non-renewable source

A finite resource, or one formed far more slowly than it is used.

Energy conservation

Energy is transferred or transformed, not created or destroyed.

Less useful energy

Energy dispersed into stores or pathways that are less useful for the intended purpose.

Pressure on a solid

Normal force per unit contact area. More area at the same force means lower average pressure.

Pressure in a still liquid

At greater depth in the same liquid, pressure due to the liquid is greater.

Equal-depth probes

In the same connected liquid at rest and under the same surface conditions, equal depth means equal pressure.

Hydraulic system

A system transmitting a pressure change through a confined liquid.

100-question practice bank

40 multiple-choice and 60 short-answer questions. This is practice, not a standardised exam. Short answers accept equivalent scientifically correct explanations. The HTML does not automatically grade your wording.

Energy forms & sources

1. Which form of energy is associated with a moving bicycle?

A. Chemical energy only

B. Nuclear energy only

C. No energy

D. Kinetic energy

2. Which source is renewable on a human timescale?

A. Petroleum

B. Uranium

C. Wind

D. Coal

3. Which source is non-renewable?

A. Flowing water

B. Coal

C. Sunlight

D. Wind

4. A raised ball has more of which energy store relative to the floor?

A. Gravitational potential

B. Sound only

C. Electrical only

D. None until it falls

5. A battery stores energy mainly in what way?

A. As stored light

B. As stored loudness

C. As wind

D. Chemically

6. When can biomass be treated as renewable?

A. Only when burned indoors

B. When forests never regrow

C. When regrowth keeps pace with use

D. Whenever any tree is cut

7. Why is uranium usually classed as non-renewable?

A. It is a kind of sunlight

B. Mined supplies are finite

C. It is a fossil fuel

D. It comes from the wind

8. Which statement about renewable energy is most accurate?

A. It can still have environmental impacts

B. It is always available everywhere

C. It never needs equipment

D. It has no limitations

9. Name two renewable energy sources.

10. Name two fossil fuels.

11. Distinguish a source from a form of energy.

12. Give an example of elastic energy storage.

13. Why is wind generation variable?

14. Why is coal not renewable for present-day human use?

15. Does a stationary raised object have zero energy? Explain.

16. Name an everyday use of sunlight.

17. Why is electricity not automatically renewable?

18. Give one possible environmental concern about an energy project.

19. What information helps a community compare two energy sources?

20. What does a sound-producing object transfer to its surroundings?

Follow the energy

21. As a ball falls from rest in the ideal no-air-resistance model, what increases?

A. Gravitational potential energy

B. Its total mechanical energy

C. Its mass

D. Kinetic energy

22. What supplies the energy in a battery torch?

A. The darkness

B. Energy created by the bulb

C. Chemical energy stored in the battery

D. Light stored in the switch

23. Which device mainly converts an electrical transfer into motion?

A. Microphone

B. Electric motor

C. Generator

D. Solar panel

24. Which device uses movement to supply an electrical transfer?

A. Generator

B. Electric heater

C. Loudspeaker

D. Kettle

25. What is a less useful output of many lamps?

A. Creation of matter

B. Destruction of all energy

C. Production of coal

D. Warming the surroundings

26. What is the main energy input to a solar photovoltaic panel?

A. Stored petrol

B. A moving handle

C. Light

D. Sound

27. What happens to energy described as wasted?

A. It is always destroyed

B. It is transferred to less useful stores or pathways

C. It disappears from existence

D. It becomes massless dust

28. Rubbing hands mainly warms them through what process?

A. Friction

B. Photosynthesis

C. Freezing

D. Nuclear fusion

29. State the conservation-of-energy idea in your own words.

30. Give the main energy chain for a torch.

31. What decreases as a ball falls towards the ground?

32. What assumption keeps mechanical energy constant in the falling-ball animation?

33. Why is a moving ball not an endless energy source?

34. Describe the energy chain in hydroelectric generation.

35. Give a useful output of an electric heater.

36. Why is heat not always a wasted output?

37. Describe a microphone's main transformation.

38. How do a motor and a generator differ?

39. What may happen to a ball's energy on impact?

40. Why should an energy diagram include the surroundings?

Pressure on solids

41. With force fixed, increasing contact area does what to pressure?

A. Increases it

B. Always doubles it

C. Changes force into mass

D. Decreases it

42. With area fixed, doubling force does what to pressure?

A. Leaves it unchanged

B. Makes it zero

C. Doubles it

D. Halves it

43. Pressure is force divided by what?

A. Temperature

B. Contact area

C. Speed

D. Mass

44. Which unit measures force?

A. Newton

B. Kilogram

C. Square metre

D. Degree Celsius

45. Why can wide bag straps feel more comfortable?

A. They remove gravity

B. They destroy the bag's mass

C. They always make the load lighter

D. They spread force over a larger area

46. A sharper cutting edge has a smaller contact area. At the same force, its pressure is what?

A. Always zero

B. Unrelated to area

C. Higher

D. Lower

47. What does normal force mean here?

A. A temperature

B. Force perpendicular to the surface

C. An average person's strength

D. Force along any direction

48. Which change lowers pressure on soft ground for the same load?

A. Using a wider support

B. Using a smaller support

C. Increasing the load

D. Standing on a narrower edge

49. Explain the difference between mass and weight.

50. What happens to pressure if area doubles while force stays fixed?

51. What happens if force and area both double?

52. Calculate pressure for 40 N on 40 square cm in the model's units.

53. Calculate pressure for 80 N on 40 square cm.

54. Calculate pressure for 40 N on 80 square cm.

55. Why can a wide foundation be useful?

56. State one assumption of the solid-pressure model.

57. Why should only one slider change in a fair comparison?

58. What does kPa stand for?

59. What is the SI unit of pressure?

60. Why is a force arrow not a measurement of pressure by itself?

Pressure in liquids

61. In the same still liquid, pressure due to the liquid does what with greater depth?

A. Decreases

B. Always stays zero

C. Depends only on container colour

D. Increases

62. Two points at equal depth in the same connected still liquid have what pressure?

A. No pressure

B. Pressure set by their labels

C. Equal pressure

D. Always different pressure

63. Depth is measured in which direction?

A. From the container label

B. Vertically below the surface

C. Along any curved wall

D. Only sideways

64. At a point in a still liquid, pressure acts in what directions?

A. All directions

B. Downwards only

C. Upwards only

D. To the left only

65. Which comparison best isolates the effect of depth?

A. Change liquid and depth together

B. Change every condition

C. Use different planet gravities

D. Use the same liquid and move a probe deeper

66. Why are deeper parts of a dam subject to greater liquid pressure?

A. Gravity stops at the surface

B. Pressure only exists in air

C. Pressure increases with depth

D. Water becomes a solid at the bottom

67. At the liquid surface, what is the modelled pressure due to the liquid above that point?

A. Negative mass

B. Zero

C. The highest reading

D. Always 100 kPa

68. Which other liquid property can affect pressure at a fixed depth?

A. Density

B. Colour alone

C. Name of the container

D. Printed logo

69. Describe what happens as the virtual probes move deeper.

70. Why do the two probes show equal pressure?

71. Does a wide tank always have greater pressure than a narrow tank at the same depth?

72. Why is 'liquid pressure only pushes down' incorrect?

73. What does the virtual pressure bar leave out?

74. Does a zero reading at the surface mean there is no atmospheric pressure?

75. Why should a depth comparison use the same liquid?

76. What does qualitative treatment mean in this lesson?

77. What is one reason the simulation is safer than an improvised pressure experiment?

78. Where is pressure due to still water greater: near the top or deeper down?

79. Why can changing container shape alone leave pressure at a point unchanged?

80. What should a learner do before a real liquid-pressure practical?

Useful science, safer choices

81. Hydraulic brakes mainly transmit pressure through what?

A. An empty vacuum

B. A sheet of paper

C. Light alone

D. A confined liquid

82. As friction brakes slow a vehicle, some kinetic energy becomes what?

A. New coal

B. A greater vehicle mass

C. Thermal energy in brakes and surroundings

D. Destroyed energy

83. Which device mainly compresses gas rather than liquid?

A. Water tank wall

B. Bicycle pump

C. Liquid-filled hydraulic line

D. Water-filled syringe in a liquid model

84. What helps push a drink up a straw?

A. Atmospheric pressure acting on the drink

B. A vacuum pulling without any pressure difference

C. Water creating energy

D. The straw destroying gravity

85. Which is a safe approach to a hydraulic-brake lesson?

A. Open a vehicle brake line

B. Test brakes on a busy road

C. Pressurise unknown containers

D. Use a simulation and teacher explanation

86. Which energy-related hazard should be avoided?

A. Using the virtual slider

B. Discussing energy sources

C. Looking directly at the Sun

D. Reading a diagram

87. Why should fuels not be burned in enclosed indoor spaces?

A. It removes all fire risk

B. Harmful gases can accumulate

C. It makes energy renewable

D. Indoor air cancels gravity

88. What is a safe rule for siphons?

A. Never start one by mouth with fuel or unknown liquids

B. Taste the liquid first

C. Use any chemical without supervision

D. Treat it as an endless energy source

89. Explain the pressure role of brake fluid.

90. What happens to some motion-related energy when brakes act?

91. How does a straw demonstrate a pressure difference?

92. How does a bicycle pump differ from a simple hydraulic system?

93. Name one safety precaution around electrical energy.

94. Name one precaution around loud sound.

95. Why is an animation not a guide to repairing brakes?

96. Why is a siphon not an endless energy source?

97. Give a safety reason to keep hands away from moving machinery.

98. How can learners share their work while protecting privacy?

99. What should a learner do if a classroom device seems hot or damaged?

100. Describe a safe community energy-learning activity.

Answer key & explanations

1. D. Kinetic energy

Motion is associated with kinetic energy.

2. C. Wind

Wind is continually produced by atmospheric processes.

3. B. Coal

Coal forms far more slowly than people use it.

4. A. Gravitational potential

Height in a gravitational field matters even before motion.

5. D. Chemically

Chemical processes in the battery can supply an electrical transfer.

6. C. When regrowth keeps pace with use

Renewability depends on replenishment and management.

7. B. Mined supplies are finite

Nuclear fuel is not a fossil fuel, but it is finite.

8. A. It can still have environmental impacts

Renewable does not mean impact-free.

9. For example, sunlight and wind.

Flowing water and geothermal resources are other examples.

10. For example, coal and petroleum.

Natural gas is also a fossil fuel.

11. A source supplies energy; a form describes how energy is stored or transferred.

Wind is a source; kinetic energy describes motion.

12. A stretched spring or elastic band.

Discuss the example without launching objects.

13. Wind speed changes with weather and location.

A renewable resource need not be constantly available.

14. Its natural formation is far slower than its consumption.

Human timescales matter in the classification.

15. No. It can have gravitational potential energy relative to a lower level.

Not all energy is kinetic.

16. For example, supplying a solar panel or helping dry clothes.

A solar panel and solar heating use different transfer pathways.

17. It can be generated using renewable or non-renewable resources.

Check the primary energy source.

18. For example, habitat disruption, air pollution or land use.

The concern depends on the type and location of the project.

19. Availability, reliability, cost, safety and environmental effects.

There is rarely a single factor to consider.

20. Energy through vibrations in a medium.

Sound requires a medium; it is not a material substance.

21. D. Kinetic energy

The loss of gravitational potential energy becomes kinetic energy.

22. C. Chemical energy stored in the battery

The battery can supply an electrical transfer.

23. B. Electric motor

Motors produce mechanical motion.

24. A. Generator

A generator is driven mechanically.

25. D. Warming the surroundings

Not all input is converted into useful light.

26. C. Light

A photovoltaic panel supplies an electrical output from light.

27. B. It is transferred to less useful stores or pathways

Energy is conserved even when dispersed.

28. A. Friction

Mechanical activity transfers energy to thermal stores.

29. Energy can be transferred or transformed, but is not created or destroyed.

Track the whole system and surroundings.

30. Chemical store in battery, electrical transfer, light and warming.

Include the less useful thermal transfer.

31. Its gravitational potential energy relative to the ground.

Its kinetic energy increases in the ideal model.

32. Air resistance and other dissipative effects are ignored before impact.

A real falling object is not perfectly isolated.

33. Its energy came from earlier transfers and becomes dispersed through interactions.

A perpetual motion source would contradict energy accounting.

34. Gravitational energy of water, motion of water and turbine, electrical transfer from a generator.

Real systems also warm their surroundings and produce sound.

35. Heating the room or intended object.

An output is useful relative to a purpose.

36. Heating can be the intended function of a device.

A heater and a lamp have different purposes.

37. Sound-related vibrations are converted into an electrical signal.

The signal can then be amplified or recorded.

38. A motor uses an electrical supply for motion; a generator uses motion for an electrical supply.

Their main transfer directions differ.

39. It can become sound, warming and deformation of the ball and surface.

The animation stops just before impact.

40. Some energy is transferred to them, so otherwise the accounting is incomplete.

Less useful transfers still conserve energy.

41. D. Decreases it

The same force is spread over more area.

42. C. Doubles it

Pressure is proportional to normal force at fixed area.

43. B. Contact area

The area must be expressed in a suitable square unit.

44. A. Newton

Mass and force are different quantities.

45. D. They spread force over a larger area

The same load can produce less pressure.

46. C. Higher

A smaller area concentrates the force.

47. B. Force perpendicular to the surface

Only the perpendicular component enters this simple pressure comparison.

48. A. Using a wider support

Increasing contact area reduces average pressure.

49. Mass is measured in kilograms; weight is a gravitational force measured in newtons.

They are related, but they are not the same quantity.

50. It halves.

Pressure is inversely related to contact area at fixed force.

51. Average pressure stays the same.

The ratio of force to area is unchanged.

52. 10 kPa.

40 divided by 40 is 1 N per square cm, equal to 10 kPa.

53. 20 kPa.

Doubling force at the same area doubles the pressure.

54. 5 kPa.

Doubling area at the same force halves pressure.

55. It spreads the building's force over more ground area.

Ground conditions and engineering design still matter.

56. For example, contact pressure is treated as uniform.

The force is also assumed perpendicular to the contact.

57. So the effect of force or area can be isolated.

Changing both at once may hide the relationship.

58. Kilopascal.

One kilopascal is one thousand pascals.

59. Pascal.

One pascal is one newton per square metre.

60. Pressure also depends on the contact area.

The same force can produce different pressures on different areas.

61. D. Increases

Deeper points support a greater overlying liquid column.

62. C. Equal pressure

At the same depth under the same conditions, liquid pressure is equal.

63. B. Vertically below the surface

Vertical depth is the relevant quantity.

64. A. All directions

Liquid pressure is not limited to one direction.

65. D. Use the same liquid and move a probe deeper

Keep other relevant factors fixed.

66. C. Pressure increases with depth

A real dam also has other engineering requirements.

67. B. Zero

The display excludes the surrounding air pressure.

68. A. Density

The model keeps liquid type fixed.

69. Both show greater pressure due to the water.

They stay at equal depth and therefore remain equal.

70. They are at the same vertical depth in the same liquid at rest.

Horizontal position alone does not change this relationship.

71. No, not for the same liquid and surface conditions at rest.

Depth and liquid properties matter, not width alone.

72. Pressure at a point acts in all directions.

Container sides also experience pressure forces.

73. The surrounding air pressure.

It displays pressure due to the liquid relative to the surface.

74. No.

A relative display is not the total absolute pressure.

75. Changing liquid density could change pressure independently of depth.

That would make the comparison less controlled.

76. Describe the trend without calculating exact liquid pressures.

The important idea is greater depth means greater pressure.

77. It avoids pressurised containers, spills or dangerous diving.

Do not drill or pressurise containers unsupervised.

78. Deeper down.

Compare vertical depths under the same conditions.

79. For the same liquid at rest, pressure depends on depth and surface conditions rather than shape alone.

The point must remain at the same vertical depth.

80. Follow the teacher's approved method and safety instructions.

A conceptual animation is not a practical procedure.

81. D. A confined liquid

Brake fluid transmits a pressure change.

82. C. Thermal energy in brakes and surroundings

The energy is transferred, not destroyed.

83. B. Bicycle pump

A pump can force compressed air into a tyre.

84. A. Atmospheric pressure acting on the drink

Lower pressure in the straw creates the needed difference.

85. D. Use a simulation and teacher explanation

Real brake systems are safety-critical.

86. C. Looking directly at the Sun

Bright light can damage eyes.

87. B. Harmful gases can accumulate

Combustion can cause poisoning as well as fires.

88. A. Never start one by mouth with fuel or unknown liquids

Use teacher-approved demonstrations or simulations.

89. It transmits a pressure change through a confined liquid to the brake mechanism.

The actual system has components not fully shown in the animation.

90. It is transferred into thermal stores of the brakes and surroundings.

It is not destroyed.

91. Reducing pressure inside allows outside air pressure to push the drink up.

Suction is not a new source of energy.

92. The pump compresses gas; the hydraulic system transmits pressure through liquid.

Liquids are much less compressible than gases.

93. Keep away from exposed mains wiring and use only approved supervised equipment.

A simulation requires no real circuit construction.

94. Avoid excessive volume and use suitable hearing protection where needed.

Sound can carry enough energy to harm hearing.

95. It omits safety-critical details and is only a conceptual model.

Vehicle work requires competent trained personnel.

96. Its flow uses gravitational and pressure differences that cannot supply unlimited energy.

Energy conservation still applies.

97. Moving parts can trap or injure hands.

Study the energy transfer without touching equipment.

98. Share an explanation or lesson link without names and personal results.

The original curriculum can be referenced by an official link.

99. Stop using it and tell the teacher without touching hazardous parts.

Do not attempt an unsupervised repair.

100. For example, compare local energy sources using observation, public information and discussion.

Do not enter power sites, climb structures or handle hazardous equipment.

Watch, share & review

Optional videos load only after a click. Titles and publishers were verified using YouTube metadata, but the complete videos have not been reviewed for classroom suitability. Teachers should check age fit, scientific accuracy, extension content and advertising. Internet is required; local-file embeds may need the direct YouTube link instead.

Conservation Of Energy | Energy | Physics | FuseSchool - FuseSchool - Global Education

https://www.youtube.com/watch?v=Um04eHOqGlM

Pressure in liquids | Matter | Physics | FuseSchool - FuseSchool - Global Education

https://www.youtube.com/watch?v=TpHv-sGE_Dk

Share the original lesson files or the website link, not the source curriculum PDF. Student completion stays in the current browser and device and is separate for each topic. There are no accounts, class dashboards or server-side student records.

Before classroom use, review the teaching text, numeric pressure conversions, qualitative liquid-pressure models, answer keys, videos, accessibility and safety. Use supervised activities to assess practical skills.

Source: KICD Grade 8 Integrated Science, revised 2024. Source links and exact page references appear on the cover.